1 Quick Review

Over these first four weeks, you got learned the methods of linear algebra that will follow you until the end of the course. For example, you've learned about inverses, matrices, vectors, linear transformations, $\mathbb{R}^n$, and determinants.

You have unlocked 7 out of the 21 (33.3%) equivalent statements you will encounter!

Theorem 1: Invertible Matrix Theorem (IMT)

If $\mathbf{A}$ is an $n \times n$ invertible (nonsingular) matrix, then the following statements are equivalent.

1. $\mathbf{A}$ is invertible.
2. $\mathbf{A}\vec{x} = \vec{0}$ has only the trivial solution.
3. The RREF of $\mathbf{A}$ is $\mathbf{I_n}$.
4. $\mathbf{A}$ can be expressed as a product of elementary matrices.
5. $\mathbf{A}\vec{x} = \vec{b}$ is consistent for every $n \times 1$ matrix $\vec{b}$.
6. $\mathbf{A}\vec{x} = \vec{b}$ has exactly one solution for every $n \times 1$ matrix $\vec{b}$.
7. $\det(\mathbf{A}) \neq 0$.

More recently you learned about how to take determinants. Below are two approaches two finding the determinant.

- You could determine the *cofactors* (C_{ij}) of the determinant and perform the summation as shown below. This is the traditional method.

$$\det(\mathbf{A}) = a_{1j}C_{1j} + a_{2j}C_{2j} + \cdots + a_{nj}C_{nj}$$

- Alternatively, you could evaluate the determinant using row operations as shown below. Although this will work it often takes much longer than cofactor expansion.

  If $\mathbf{A}$ is the original $n \times n$ matrix and $\mathbf{B}$ is the resultant matrix (after you perform the operation) than the determinants are related by:

  - $\det(\mathbf{B}) = k\det(\mathbf{A}),\ k \in \mathbb{R}^n$ for a **scaling** operation.
  - $\det(\mathbf{B}) = -\det(\mathbf{A})$ for an **interchange** operation.
  - $\det(\mathbf{B}) = \det(\mathbf{A})$ for a **replacement** operation (adding a scalar multiple of one row to another row).

2 Problems

1. Given the matrices **A**, **B**, and **C**:

$$\mathbf{A} = \begin{bmatrix} 2 & 7 & 0 & 2 \\ 2 & 0 & 0 & 6 \\ 5 & 3 & 0 & 7 \\ 1 & 0 & 10 & 3 \end{bmatrix}, \mathbf{B} = \begin{bmatrix} 1 & 2 & -1 \\ 2 & 0 & 2 \\ 1 & 5 & 3 \end{bmatrix}, \mathbf{C} = \begin{bmatrix} 0 & 0 & 6 \\ 3 & 0 & 4 \\ 0 & 10 & 3 \end{bmatrix},$$

solve the following problems.

(a) Find $\mathbf{A}^T$.

(b) Find $\det(\mathbf{A})$.

(c) Find $\mathbf{B}^{-1}$.

(d) Find $\det(\mathbf{B}^{-1})$.

(e) Find $\det(\mathbf{BC})$.

2. Create two matrices $\mathbf{A}$ and $\mathbf{B}$ that satisfy the following condition: $\mathbf{AB} = \mathbf{BA}$, where $\mathbf{A} \neq \mathbf{B} \neq \mathbf{I}$.

3. **(Proof).** Suppose there are two diagonal matrices $\mathbf{C}$ and $\mathbf{D}$. Prove that the matrix multiplication between them *is* commutative.

4. Let $T : \mathbb{R}^3 \to \mathbb{R}^3$ be defined by the matrix $\mathbf{A}$:

$$\mathbf{A} = \begin{bmatrix} 2 & -4 & 5 \\ 2 & 0 & 1 \\ 0 & 5 & 2 \end{bmatrix}$$

   (a) Specify the domain and codomain.
      i. Domain:
      ii. Codomain:
   (b) What would be the size of the matrix if it were to have a domain of $\mathbb{R}^5$ and a codomain of $\mathbb{R}^2$?

$$\_\_\_ \times \_\_\_$$

   (c) Write the transformation in the $T(input) = (output)$ format.

   (d) What are the *image* and *pre-image* in a transformation?

   (e) Find the image of $\vec{x} = (1,\ 4,\ 5)$ under the transformation T.

5. Use Cramer's Rule to solve $\mathbf{A}\vec{x} = \vec{b}$, where

$$\mathbf{A} = \begin{bmatrix} 1 & 2 & -1 \\ 2 & 0 & 2 \\ 1 & 5 & 3 \end{bmatrix} \text{ and } \vec{b} = \begin{bmatrix} 1 \\ 3 \\ 5 \end{bmatrix}.$$